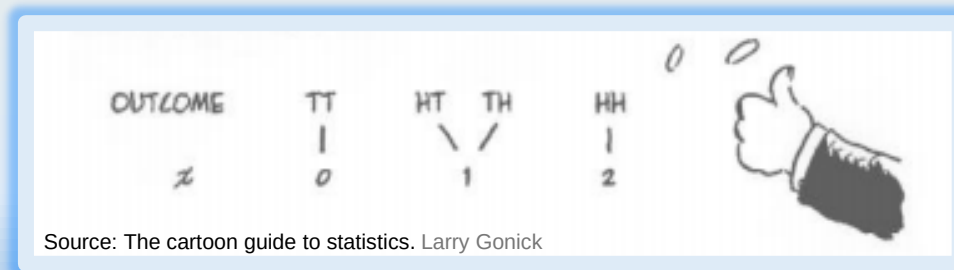


Mathematics II (part 1)

DISCRETE PROBABILITY DISTRIBUTION



DISCRETE PROBABILITY DISTRIBUTION

INTRODUCTION

What are we going to learn?

Understanding Uncertainties

- Random Variables
- Discrete Random Variables

Probability distributions

- Mean (expected value)
- Variance
- Standard Deviation

Binomial Random Variables

- Identification
- Mean (expected value)
- Variance

UNDERSTANDING UNCERTAINTIES

RANDOM VARIABLES

Understanding uncertainty

Random Variables

A random variable, usually written as a capital letter- e.g. X , is a variable whose possible values are numerical outcomes of a random phenomenon.

Example: To describe the probabilities of the outcomes when tossing a dice, we use:

Let X_1 be the event of getting a '1', then $P(X_1) = \frac{1}{6}$.

Let X_2 be the event of getting a '2', then $P(X_2) = \frac{1}{6}$.



Alternatively, let X be the **number** obtained when tossing a dice, then

$$P(X = 1) = \frac{1}{6} , \quad P(X = 2) = \frac{1}{6} , \quad \dots$$

In other words, mapping outcomes from random process to numbers.

Understanding uncertainty

Random Variables

A random variable, usually written as a capital letter- e.g. X , is a variable whose possible values are numerical outcomes of a random phenomenon.

Another example: flipping two coins, Let X be number of heads you get.

*Random
Variable*

X

=

*Possible
Values*

0

1

2

*Random
Events*



{TT}

{HT, TH}

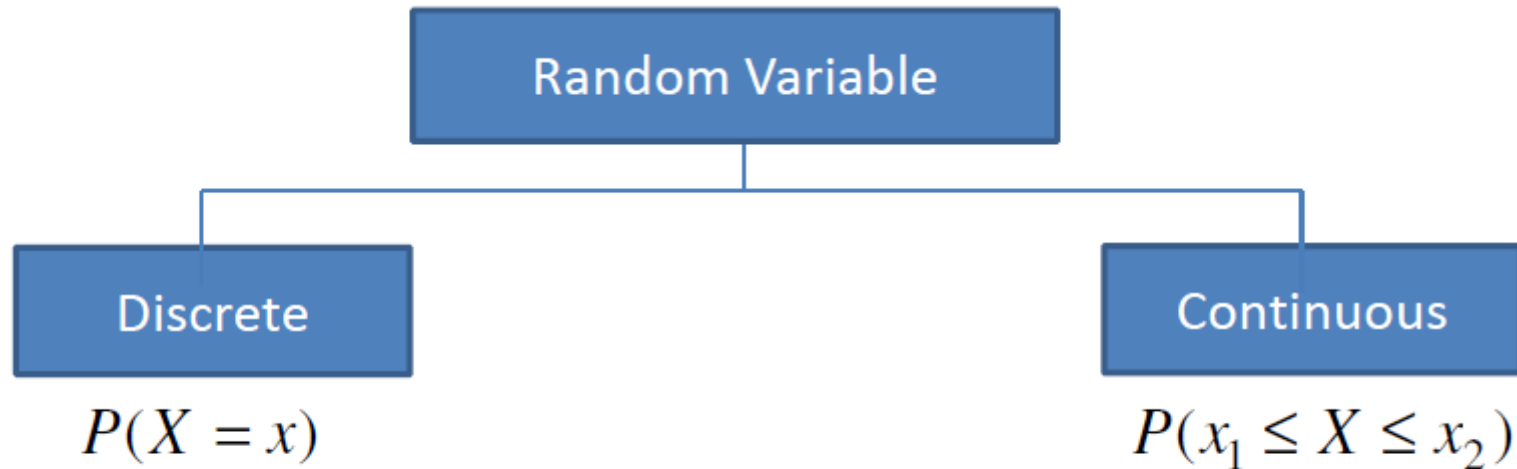
{HH}

Difference between a **variable** and a **random variable**

- A **variable** is a symbol that represents some **quantity**. There are no assumptions about its value
- A **random variable** is a value that follows some **probability distribution**. In other words, it's a value that is subject to **some randomness or chance**.

Understanding uncertainty

Types of random variables



By convention, we use capital letter, e.g. X to represent **random variable**, and small letter, e.g. x to represent the **value**.

Understanding uncertainty

Types of random variables

- A **Discrete Random Variable** can take one of a countable list of distinct values. For discrete random variables, we can find probabilities for **exact outcomes**.
- A **Continuous Random Variable** can take any value in an interval or collection of intervals. For continuous variables we are limited to finding probabilities for **intervals of values**.

UNDERSTANDING UNCERTAINTIES

DISCRETE RANDOM VARIABLES

Choose all that apply as discrete random variables

- Y = the year in which a student from this class is born
- Z = number of babies born tomorrow in the universe
- X = exact winning time for swimming a sprint
- W = exact winning time for swimming a sprint rounded to nearest tenth

Probability of discrete random variables

- Let X = number of heads after 3 flips of a fair coin
- Possible outcomes: HHH, HTH, HHT, HTT, TTT, THT, TTH, THH
- Possible value of X = 0,1,2,3
 - $P(X=0) = ?$
 - $P(X=1) = ?$
 - $P(X=2) = ?$
 - $P\{X=3\} = ?$

Probability of discrete random variables

- Let X = number of heads after 3 flips of a fair coin
- Possible outcomes: HHH, HTH, HHT, HTT, TTT, THT, TTH, THH
- Possible value of X = 0,1,2,3
 - $P(X=0) = ?$
 - $P(X=1) = ?$
 - $P(X=2) = 3/8$
 - $P\{X=3\} = ?$

Probability of discrete random variables

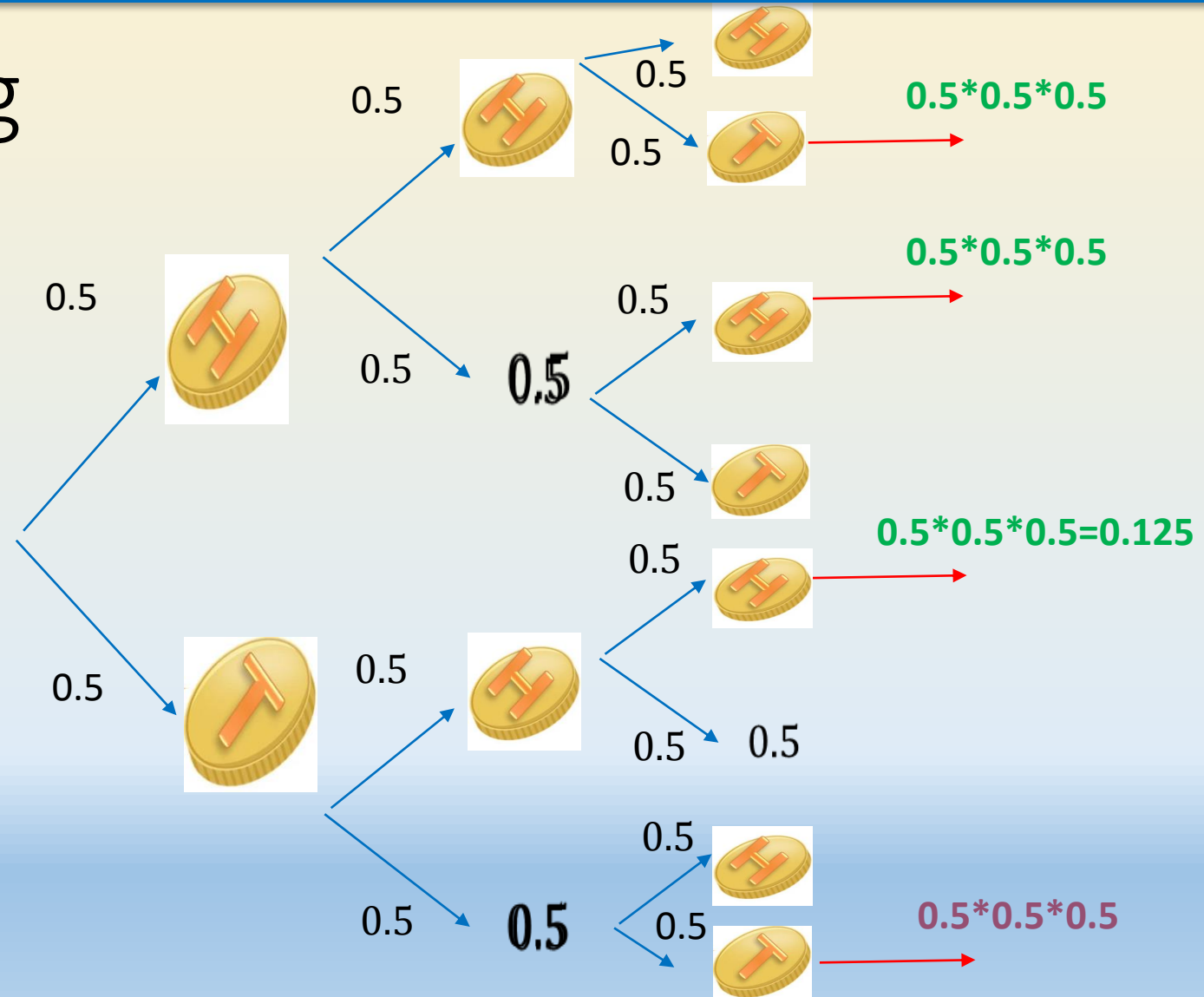
- Let X = number of heads after 3 flips of a fair coin
- Possible outcomes: HHH, HTH, HHT, HTT, TTT, THT, TTH, THH
- Possible value of X = 0,1,2,3
 - $P(X=0) = 1/8$
 - $P(X=1) = ?$
 - $P(X=2) = 3/8$
 - $P\{X=3\} = ?$

Understanding uncertainty

Discrete random variables: Probability Calculations

Another way for finding probability

- Let X = number of heads after 3 flips of a fair coin
- Possible value of X = 0, 1, 2, 3
 - $P(X=0) = 0.5 * 0.5 * 0.5 = 1/8$
 - $P(X=1) = ?$
 - $P(X=2) = 0.125 + 0.125 + 0.125 = 3/8$
 - $P\{X=3\} = ?$



PROBABILITY DISTRIBUTION

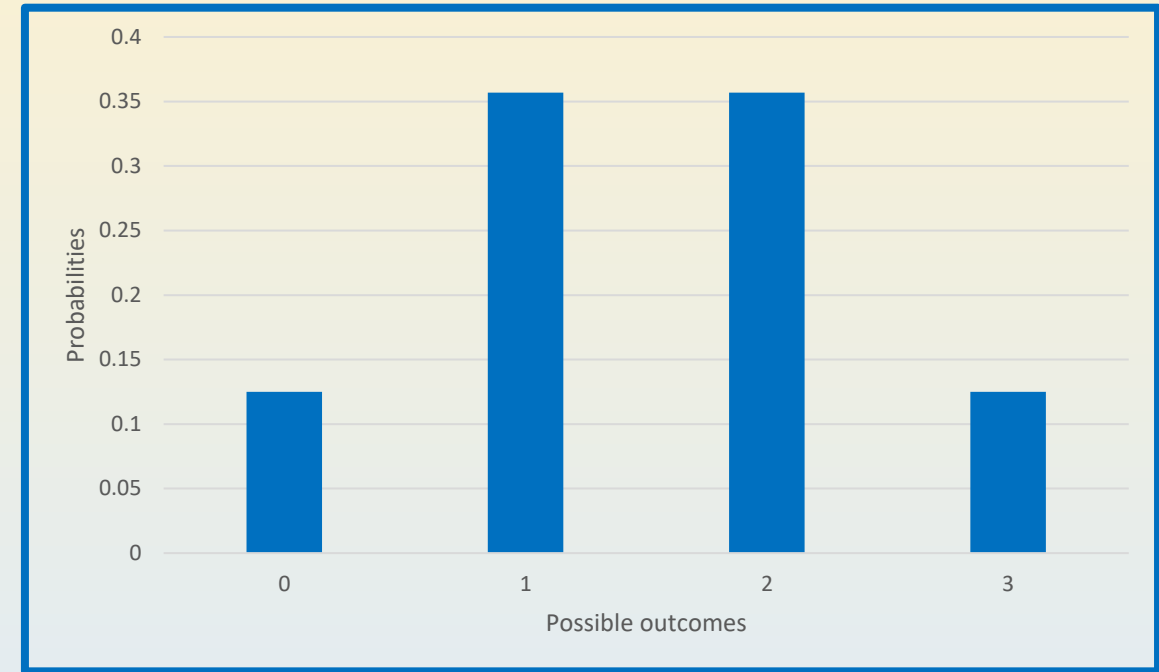
EXPECTED VALUES and MEASURING SPREAD

Probability distributions

Introduction

- Let X = number of heads after 3 flips of a fair coin
- Possible value of X = 0,1,2,3
 - $P(X=0) = ?$
 - $P(X=1) = ?$
 - $P(X=2) = ?$
 - $P(X=3) = ?$

Definition: The probability distribution of a discrete random variable X is the collection of all possible values x with their associated probabilities. It is usually shown as a table, a graph or a formula.



We have just constructed a probability distribution for X

Discrete Probability Distribution for Random variable X

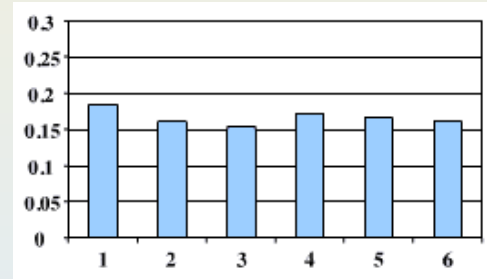
Probability distributions

Frequency vs Probability distribution

- **Frequency** Distribution

Discrete **data variables** can be represented by frequency distribution

Class	Frequency	Relative Frequency
1	18	0.18
2	16	0.16
3	15	0.15
4	18	0.18
5	17	0.17
6	16	0.16

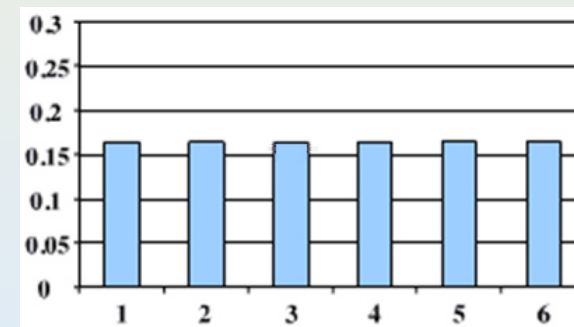


$$0 \leq \frac{f_i}{n} \leq 1 \quad \text{and} \quad \sum \frac{f_i}{n} = 1$$

- **Probability** Distribution

Discrete **random variables** are described by probability distributions.

x	1	2	3	4	5	6
P(X=x)	1/6	1/6	1/6	1/6	1/6	1/6



$$0 \leq P(X = x) \leq 1 \quad \text{and} \quad \sum P(X = x) = 1$$

Probability distributions

Expected Value (aka Mean)

- Mean

For a Frequency Distribution, we discussed how to get the mean

Mean in Descriptive Statistics

$$\bar{x} = \frac{\sum x_i f_i}{n} = \sum x_i \frac{f_i}{n}$$

- Expected Value (probability distribution)

For a random variable X , we want to calculate the expected value.

Definition: The mean or expected value of a discrete random variable X is:

$$E(X) = \mu = \sum xP(X = x)$$

Example: Let X be the number obtained when tossing a dice.

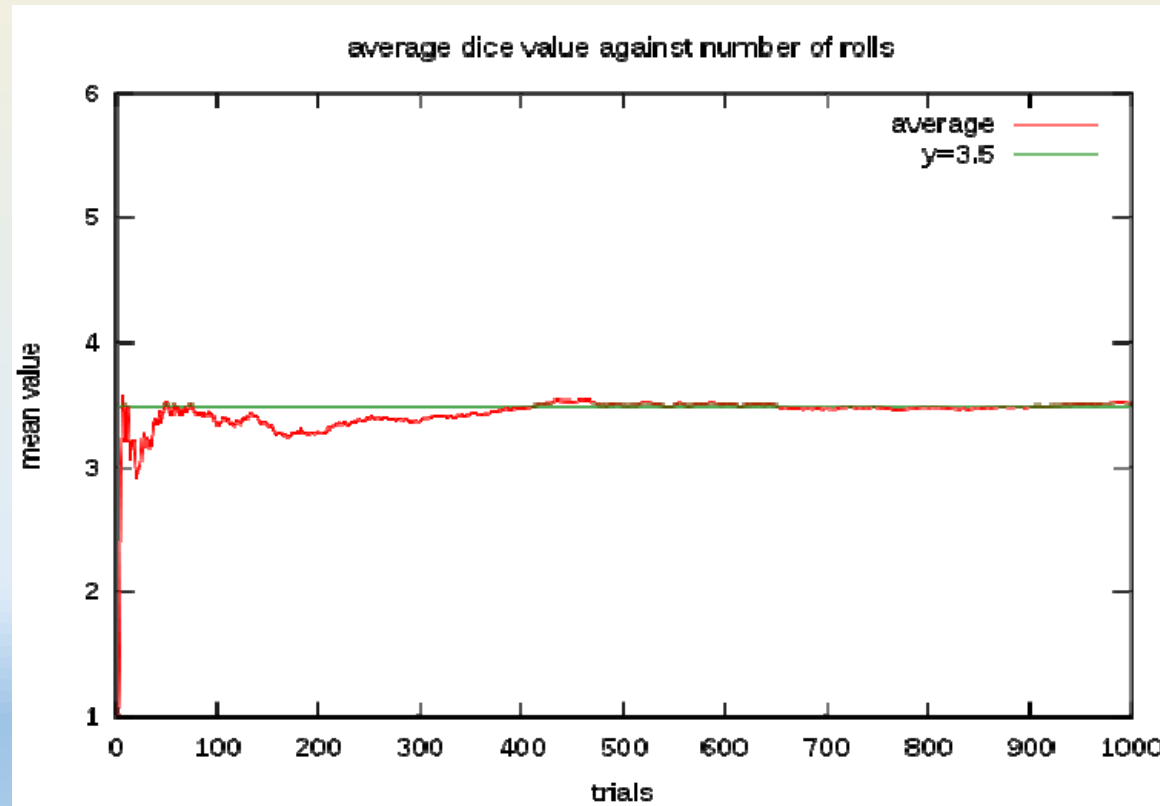
The mean or expected value of X is:

$$\begin{aligned} E(X) &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= 3.5 \end{aligned}$$

Probability distributions

Law of large numbers

Expected value is of practical importance because it has been shown that **as the number of trials** increases, the **mean** outcome will tend to become equal to the **expected value**, also called **Law of Large Numbers**.



Probability distributions

Exercise: Question

Darren is baking 2 batches of cookies. Since he tends to be quite forgetful, there's a good chance he might burn the cookies, and then he can't sell them. Each batch is independent, and the probability of his first batch being not burned (sell-able) is 50%, and the probability of his second batch being not burned (sell-able) is 70%.

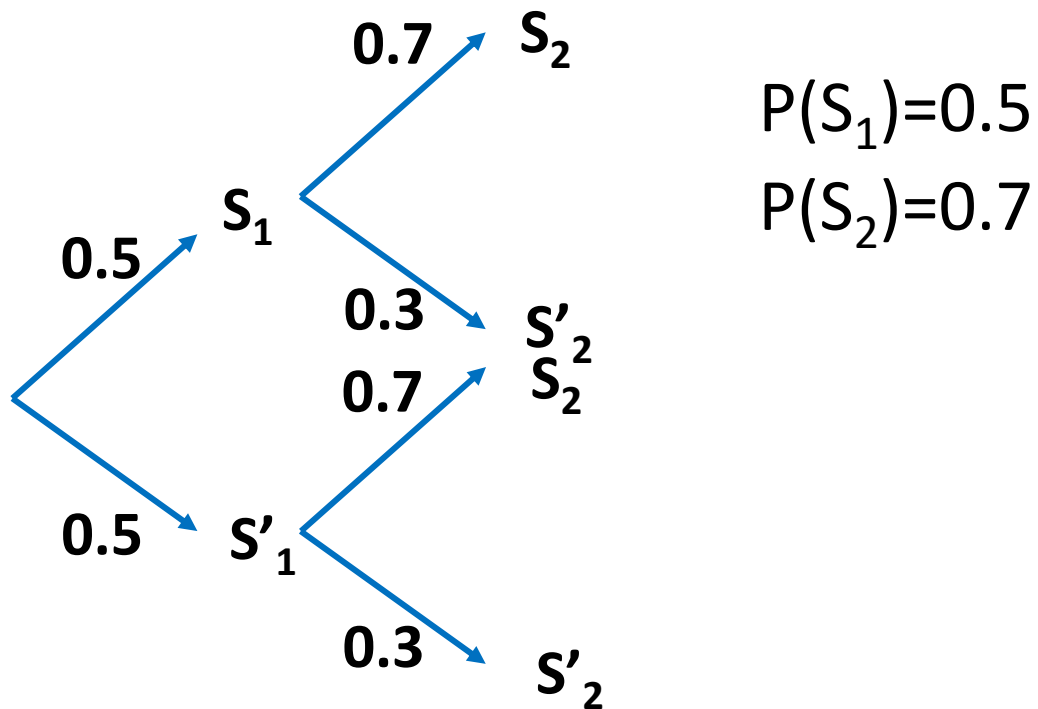
- If **X** is a random variable that represents the **number of sellable batches** of cookies Darren makes,
- Construct a probability distribution of **X**

Probability distributions

Exercise: Solution

Let X = the number of sellable batches of cookies Darren makes

- Possible value of $X = 0, 1, 2$
- Possible outcomes: (S_1, S_2) , (S_1, S'_2) , (S'_1, S_2) , (S'_1, S'_2)



Probability distributions

Exercise: Solution

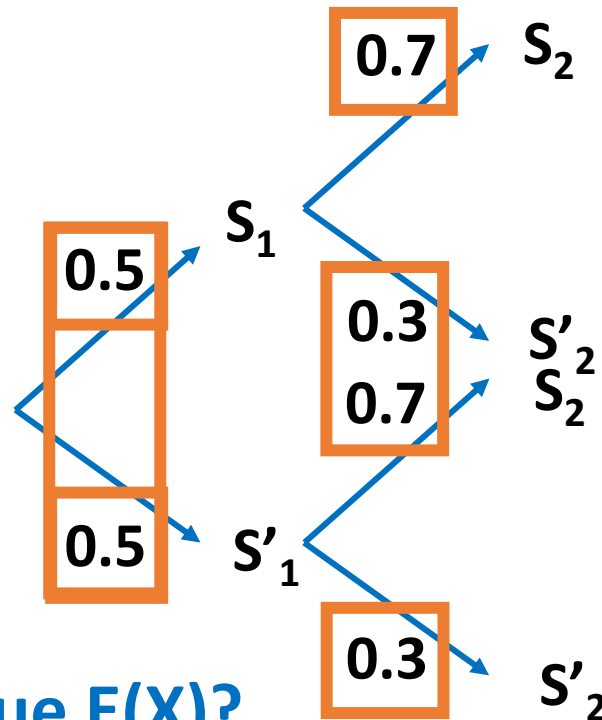
Let X = the number of sellable batches of cookies Darren makes

- Possible value of $X = 0, 1, 2$
- Possible outcomes: (S_1, S_2) , (S_1, S'_2) , (S'_1, S_2) , (S'_1, S'_2)

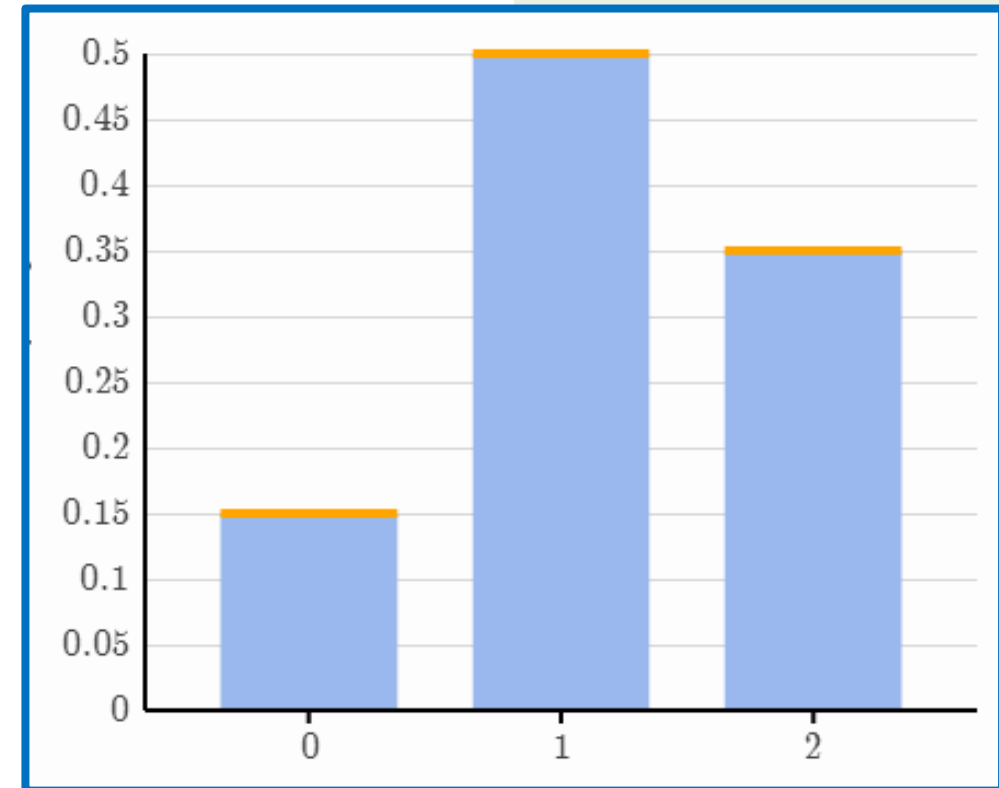
$$P(X=0) = 0.5 \cdot 0.3 = 0.15$$

$$P(X=1) = 0.5 \cdot 0.7 + 0.5 \cdot 0.3 = 0.5$$

$$P(X=2) = 0.5 \cdot 0.7 = 0.35$$



What is the expected value $E(X)$?



Probability distributions

Exercise: Solution for Expected value $E(x)$

Let X = the number of sellable batches of cookies Darren makes

- Possible value of $X = 0, 1, 2$
- Possible outcomes: (S_1, S_2) , (S_1, S'_2) , (S'_1, S_2) , (S'_1, S'_2)

What is the expected value $E(X)$?

$$P(X=0) = 0.5 * 0.3 = 0.15$$

$$P(X=1) = 0.5 * 0.7 + 0.5 * 0.3 = 0.5$$

$$P(X=2) = 0.5 * 0.7 = 0.35$$

$$E(X) = 0 * 0.15 + 1 * 0.5 + 2 * 0.35 = 1.2$$

Multiply each value by its probability

$$E(X) = \sum x \bullet P(X = x)$$

$E(X)$ is the expectation of X

After multiplying each value pair, add the whole lot together

Probability distributions

Measuring spread (revisited)

In Descriptive Statistics, numeric variable can be described using **variance** and **standard deviation**. Similarly, **discrete random variables** also have a variance and standard deviation.

Descriptive Statistics (relative frequencies)

Variance and standard deviation in descriptive statistics

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Probability distribution

The **variance** of a discrete **random variable** X is:

$$\text{Var}(X) = \sigma^2 = \sum (x - \mu)^2 P(X = x)$$

The **standard deviation** of a discrete **random variable** X is:

$$\sigma = \sqrt{\sigma^2}$$

Probability distributions

Measuring spread (revisited), Exercise: question

Assume that you have \$100 to invest in one of the two investment plans with different potential gains and probabilities:

- Which plan do you invest in?

Plan 1		Plan 2	
$X = \text{Net Gain}$	Probability	$Y = \text{Net Gain}$	Probability
\$5,000	.001	\$20	.3
\$1,000	.005	\$10	.2
\$0	.994	\$4	.5

Probability distributions

Measuring spread (revisited), Exercise: Solution (first get the expected value)

Solution: Let X be the potential gain of Plan 1, and
 Y be the potential gain of Plan 2.

$$E(X) = \mu_X = (\$5000)(0.001) + (\$1000)(0.005) + (\$0)(0.994) = \$10$$

$$E(Y) = \mu_Y = (\$20)(0.3) + (\$10)(0.2) + (\$4)(0.5) = \$10$$

Both seems equally good (or lousy).

Probability distributions

Measuring spread (revisited), Exercise: Solution (Coefficient of variance)

Coefficient of variance

$(X - \mu)^2$	p	$(X - \mu)^2 p$	$(Y - \mu)^2$	p	$(Y - \mu)^2 p$
$(\$5000 - \$10)^2 = \$24,900,100$	.001	\$24,900.10	$(\$20 - \$10)^2 = \$100.00$	.3	\$30.00
$(\$1000 - \$10)^2 = \$980,100$	.005	\$4,900.50	$(\$10 - \$10)^2 = \$0.00$	.2	\$0.00
$(\$0 - \$10)^2 = \$100$	.994	\$99.40	$(\$4 - \$10)^2 = \$36.00$	.5	\$18.00

$$\begin{aligned}
 Var(X) &= \sum (x - \mu_X)^2 P(X = x) \\
 &= \$24,900.10 + \$4,900.50 \\
 &\quad + \$99.40 \\
 &= \$29,900
 \end{aligned}$$

$$\sigma_X = \sqrt{29,900} = \$172.92$$

$$\begin{aligned}
 Var(Y) &= \sum (y - \mu_Y)^2 P(Y = y) \\
 &= \$30 + \$0 + \$18 \\
 &= \$48
 \end{aligned}$$

$$\sigma_Y = \sqrt{48} = \$6.93$$

Probability distributions

Measuring spread (revisited), Exercise: Solution (Coefficient of variance)

Coefficient of variance

$(X - \mu)^2$	p	$(X - \mu)^2 p$	$(Y - \mu)^2$	p	$(Y - \mu)^2 p$
$(\$5000 - \$10)^2 = \$24,900,100$	.001	\$24,900.10	$(\$20 - \$10)^2 = \$100.00$	.3	\$30.00
$(\$1000 - \$10)^2 = \$980,100$	.005	\$4,900.50	$(\$10 - \$10)^2 = \$0.00$	.2	\$0.00
$(\$0 - \$10)^2 = \$100$	.994	\$99.40	$(\$4 - \$10)^2 = \$36.00$	.5	\$18.00

$$\begin{aligned}
 Var(X) &= \sum (x - \mu_X)^2 P(X = x) \\
 &= \$24,900.10 + \$4,900.50 \\
 &\quad + \$99.40 \\
 &= \$29,900
 \end{aligned}$$

$$\sigma_X = \sqrt{29,900} = \$172.92$$

$$CV_X = \frac{\sigma_X}{\mu_X} = \frac{\$172.92}{\$10} = 17.29$$

$$\begin{aligned}
 Var(Y) &= \sum (y - \mu_Y)^2 P(Y = y) \\
 &= \$30 + \$0 + \$18 \\
 &= \$48
 \end{aligned}$$

$$\sigma_Y = \sqrt{48} = \$6.93$$

$$CV_Y = \frac{\sigma_Y}{\mu_Y} = \frac{\$6.93}{\$10} = 0.69$$

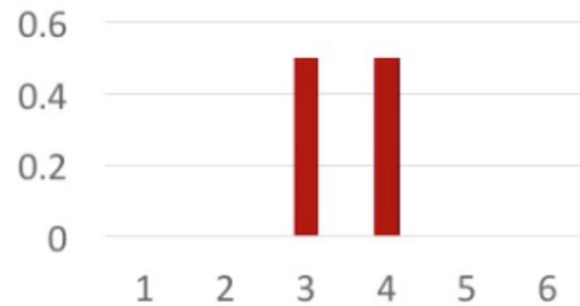
Similar as Descriptive Statistics, **coefficient of variation CV** can be applied. So we conclude that Plan 1 has higher investment risk than Plan 2.

Probability distributions

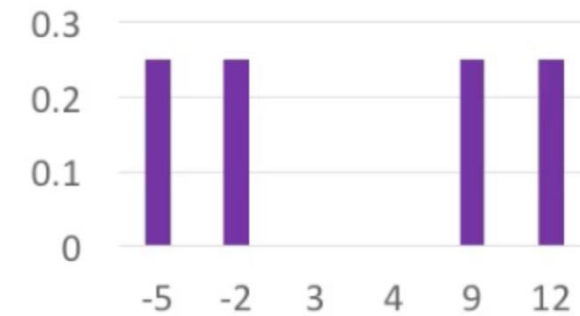
Theory questions

Which of the graphs shows the lowest and highest variance in the data?

Identical Expected Values of 3.5



$P(D=d)$

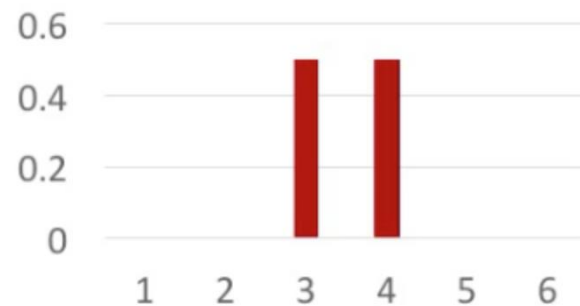


Probability distributions

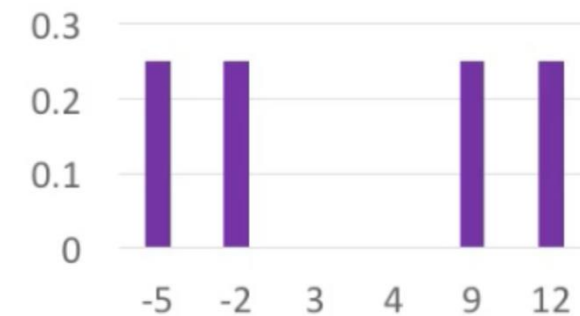
Theory questions

Which of the graphs shows the lowest and highest variance in the data?

Identical Expected Values of 3.5



$P(D=d)$



Lowest variance, with equal x-axis data, and they're close to $E(x)$

Highest variance. x-axis data vary and are away from the $E(x)$

Probability distributions

Summary

To summarize the similarity between **variables** in Descriptive Statistics and **random variables** in Probability Theory:

Empirical (Data)	Theoretical (Mathematical)
(a) Relative Frequency $\hat{p} = \frac{f_i}{n}$	(a) Probability $P(X = x_i) = p_i$
(b) Sum of all relative frequencies $\sum \frac{f_i}{n} = 1$	(b) Sum of all probabilities $\sum P(X = x_i) = 1$
(c) Mean $\bar{x} = \sum x_i \frac{f_i}{n}$	(c) Expectation or mean $E(X) = \sum x_i P(X = x_i)$
(d) Variance $s^2 = \sum (x_i - \bar{x})^2 \frac{f_i}{n-1}$	(d) Variance $Var(X) = \sum (x_i - \mu)^2 P(X = x_i)$

DISCRETE PROBABILITY DISTRIBUTION

BINOMIAL RANDOM VARIABLES

Binomial random variables

Introduction

A **binomial random variable** is a special class of random variables

In general, a random phenomenon can be modelled using **binomial random variable** if it exhibits the following 4 features:

How to identify:

1. Finite number of identical trials (for example, each coin flip, etc.)
2. Each trial must have one of two discrete outcomes (success or failure)
3. Probability of success and failure is fixed over all trials
4. Fixed number of independent trials

*For example, X = number of heads I get after **10** flips*

Binomial random variables

Binomial distribution features

- The random phenomenon consists of n identical trials
e.g. When rolling a pair of dice n times,
each roll is identical

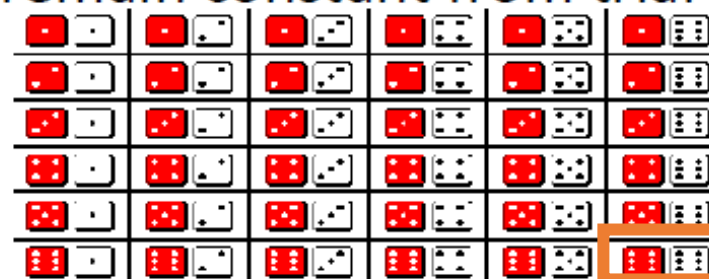


- Each trial has **two** possible outcomes, called 'success' and 'failure'
e.g. If you are interested in the outcome (6,6),
then define 'success' = (6,6), 'failure' = others



- $P(\text{'success'}) = p$ and $P(\text{'failure'}) = 1 - p$ remain constant from trial to trial; e.g. from above:

$$P(\text{'success'}) = \frac{1}{36} \quad P(\text{'failure'}) = \frac{35}{36}$$



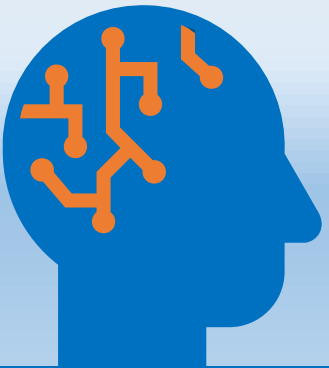
- The outcomes are **independent** from one trial to the next
e.g. When rolling a pair of dice, outcome of next roll does not depend on previous outcome.

Binomial random variables

Questions: Which of the following are Binomial distributions?

1. Let Y = number of kings I get taking two cards (one after the other) without replacement. (probability changes each time I take the cards)
2. Take a random sample of 30 sales calls, and let L equal the number of calls that result in a sale.
3. Roll 5 fair dice at once & let Z = the number of dice that land showing an even value (2, 4, or 6)
4. Take a random sample of 3 sales calls and let K = the amount from sales made—in dollars—that result from the calls. (not two outcomes success and failure, a value is expected and can be anything)
5. Roll 5 fair dice at once and let Y = the number of dice that land showing a "4".

WHY IS IT USEFUL TO
IDENTIFY A PROBLEM
AS **BINOMIAL**?



Binomial random variables

Binomial distributions: can be used to calculate any probability distribution (normal)

What is the **mean** and **variance** of obtaining heads when a fair coin is tossed 3 times?

Identify the random variable:

Let X be the number of heads obtained when the fair coin is tossed 3 times.

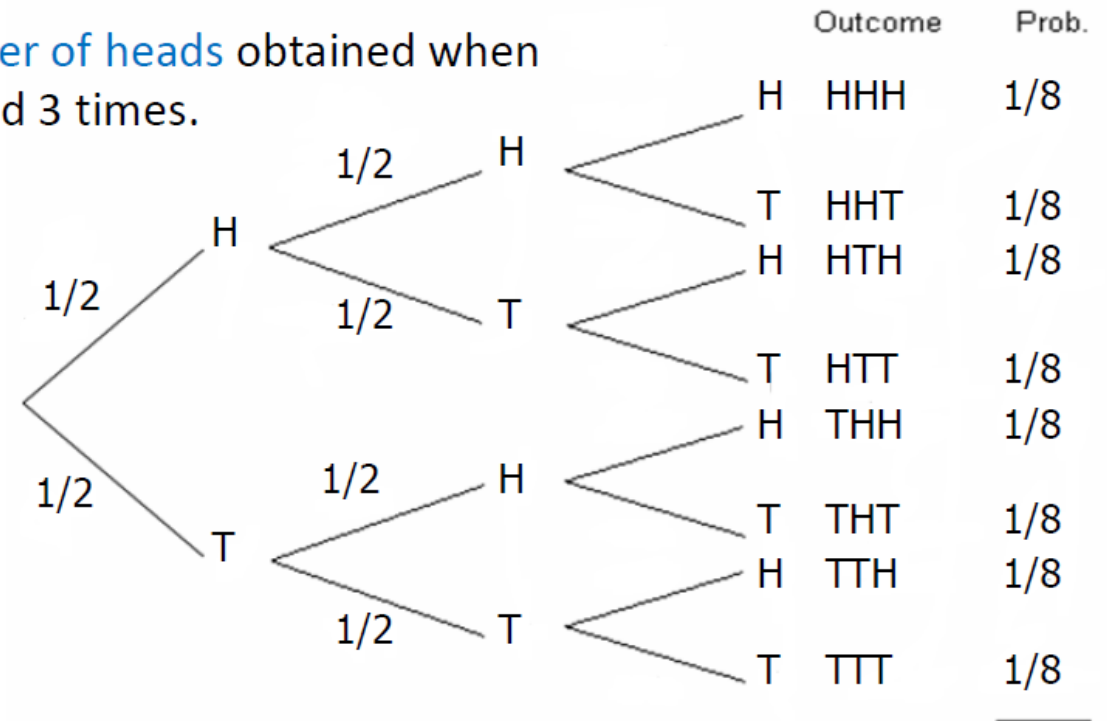
Construct a probability distribution:

Solution:

Let X be the **number of heads** obtained when the fair coin is tossed 3 times.

The **probability distribution** of X is:

x	$P(X=x)$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$



Binomial random variables

Binomial distributions: can be used to calculate any probability distribution (normal)

To calculate mean and variance:

mean:

$$\begin{aligned}\mu &= \sum xP(X = x) \\ &= 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = \frac{12}{8} = 1.5\end{aligned}$$

variance:

$$\begin{aligned}\sigma^2 &= \sum (x - \mu)^2 P(X = x) \\ &= (-1.5)^2\left(\frac{1}{8}\right) + (-0.5)^2\left(\frac{3}{8}\right) + (0.5)^2\left(\frac{3}{8}\right) + (1.5)^2\left(\frac{1}{8}\right) = 0.75\end{aligned}$$

x	(x-μ) ²	x ²	P(X=x)
0	(-1.5) ²	0	1/8
1	(-0.5) ²	1	3/8
2	(0.5) ²	4	3/8
3	(1.5) ²	9	1/8

DO WE **NEED** THESE
CALCULATIONS IF WE
IDENTIFY A **BINOMIAL**
DISTRIBUTION?



Binomial random variables

Binomial distributions

The probability distribution of a binomial random variable X is usually written as:

$$X \sim B(n, p)$$

n number of
trials

p probability of
success

Calculating the probability is straightforward through an equation:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

number of success

Combinations

where

$$\binom{n}{x} = C_x^n = \frac{n!}{(n-x)!x!}$$

n choose x
can be
calculated
using this

Binomial random variables

Binomial distributions

The probability distribution of a binomial random variable X , written as $X \sim B(n,p)$, is:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

number of success

where $\binom{n}{x} = C_x^n = \frac{n!}{(n-x)!x!}$

The mean and variance of $X \sim B(n,p)$ are:

$$E(X) = np$$

Mean, Expected value

$$Var(X) = np(1-p)$$

Variance

All calculations that were done before are replaced by these equations only!

Binomial random variables

Solving using Binomial distributions

Let X be the number of heads obtained when the fair coin is tossed 3 times. What is the **mean** and **variance**?

We can use **binomial random variable** to **model** it because:

- $n = 3$ identical trials
- 2 possible outcomes – head or tail
- $P(\text{'success'}) = P(\text{'head'}) = p = 0.5$ since we are interested in heads
- Every trial is independent

So, let X be the number of heads obtained, $X \sim B(3, 0.5)$

mean $E(X) = np = 3(0.5) = 1.5$

variance $Var(X) = np(1 - p) = 3(0.5)(1 - 0.5)$
 $= 0.75$

x	$(x-\mu)^2$	x^2	$P(X=x)$
0	$(-1.5)^2$	0	1/8
1	$(-0.5)^2$	1	3/8
2	$(0.5)^2$	4	3/8
3	$(1.5)^2$	9	1/8

mean:
 $\mu = \sum xP(X=x)$
 $= 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = \frac{12}{8} = 1.5$

variance:
 $\sigma^2 = \sum (x-\mu)^2 P(X=x)$
 $= (-1.5)^2\left(\frac{1}{8}\right) + (-0.5)^2\left(\frac{3}{8}\right) + (0.5)^2\left(\frac{3}{8}\right) + (1.5)^2\left(\frac{1}{8}\right) = 0.75$

The **mean** and **variance** of $X \sim B(n, p)$ are:

$$E(X) = np$$

Mean, Expected value

$$Var(X) = np(1 - p)$$

Variance

Binomial random variables

Solving using Binomial distributions: Exact values needed?

Do you want to know the probability when the number of heads is exactly 1 or 2 or 3? $X \sim B(3, 0.5)$

$$P(X=1) = \frac{3!}{(3-1)!1!} 0.5^1 (0.5)^{3-1} = 0.375$$

$$P(X=2) = \frac{3!}{(3-2)!2!} 0.5^2 (0.5)^{3-2} = 0.375$$

$$P(X=3) = \frac{3!}{(3-3)!3!} 0.5^3 (0.5)^{3-3} = 0.125$$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

where $\binom{n}{x} = C_x^n$

$$= \frac{n!}{(n-x)!x!}$$

The probability distribution of X is:

x	P(X=x)
0	1/8
1	3/8
2	3/8
3	1/8

BINOMIAL RANDOM VARIABLES

BINOMIAL DISTRIBUTIONS EXAMPLES

Binomial distributions examples

Solving using Binomial distributions: Real world example (i)

A quality control system selects a sample of 10 items from each batch of products for testing. If 2 or more of the sample are defective the whole batch is rejected.

If the probability of an item being defective is 0.05, what is the probability:

i. of having 2 defects in the sample?

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Binomial distributions examples

Solving using Binomial distributions: Real world example, Solution (i)

Let X be the number of defects in a batch, $X \sim B(10, 0.05)$

$$\begin{aligned} \text{(i)} \quad P(X = 2) &= \binom{10}{2} (0.05)^2 (1 - 0.05)^8 \\ &= 0.0746 \end{aligned}$$

Binomial distributions examples

Solving using Binomial distributions: Real world example (ii)

A quality control system selects a sample of 10 items from each batch of products for testing. If 2 or more of the sample are defective the whole batch is rejected.

If the probability of an item being defective is 0.05, what is the probability:

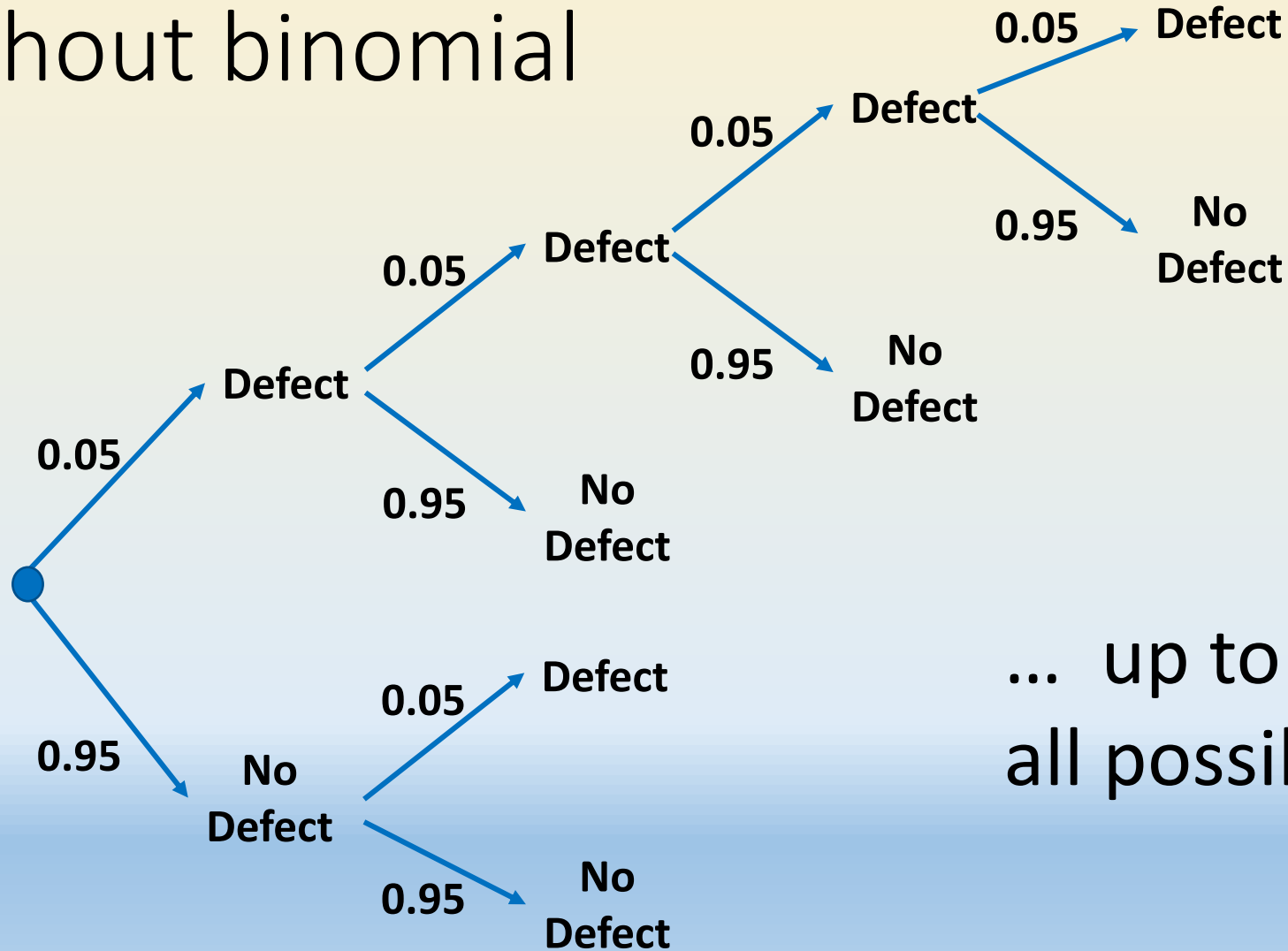
- i. of having 2 defects in the sample?
- ii. that a batch is rejected?

$$P(\text{reject}) = P(X \geq 2)$$

Binomial random variables

Solving using Binomial distributions: Real world example, Solution (ii)

Without binomial



... up to **10 times** and take
all possible combinations

Binomial random variables

Solving using Binomial distributions: Real world example, Solution (ii)

- $X \sim B(10, 0.95)$ consider the chance of success now

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

To solve this problem, what do you need to find using the above?

$$P(\text{not rejected}) = P(X = 10) + P(X = 9)$$

$$= \binom{10}{10} 0.95^{10} (1 - 0.95)^{10-10} + \binom{10}{9} 0.95^9 (1 - 0.95)^{10-9}$$

$$= (1 * 0.95^{10} * 1) + (10 * 0.95^9 * 0.05)$$

$$= (0.59874) + (0.31512)$$

$$= 0.91386$$

$$P(\text{not rejected}) = 1 - P(\text{rejected})$$

$$P(\text{rejected}) = 0.08614 \text{ (~to 5 decimal places)}$$

Binomial distributions examples

Solving using Binomial distributions: Real world examples (iii) and (iv)

A quality control system selects a sample of 10 items from each batch of products for testing. If 2 or more of the sample are defective the whole batch is rejected.

If the probability of an item being defective is 0.05, what is the probability:

- i. of having 2 defects in the sample?
- ii. that a batch is rejected?
- iii. mean number of defects in a batch?
- iv. standard deviation of number of defects in a batch?

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- mean $\mu = np$
- variance $\sigma^2 = np(1 - p)$

Binomial distributions examples

Solving using Binomial distributions: Real world examples, Solutions (iii) and (iv)

Solution: (continue)

(iii) mean number of defects in a batch

$$\begin{aligned}\mu &= np \\ &= 10 \times 0.05 = 0.5\end{aligned}$$

(iv) variance $\sigma^2 = np(1 - p)$

$$= 10 \times 0.05 \times (1 - 0.05) = 0.475$$

standard deviation of number of defects in a batch

$$\begin{aligned}\sigma &= \sqrt{\sigma^2} \\ &= \sqrt{0.475} = 0.689\end{aligned}$$

Binomial distributions examples

Solving using Binomial distributions: Over Booking example

A particular flight has room for 30 passengers. The airline knows from past experience that, on average, only 80% of all passengers holding reservations show up for the flight.

Suppose the airline sells 32 tickets for this flight. What is the probability that the flight is overbooked, i.e., that more people show up for the flight than there are seats available?

Assume we want to solve this question using Binomial, what would be the random variable X ?

- $X \sim B(32, 0.8)$ which means $n = 32$ and $p = 0.8$

To solve this problem, what do you need to find using the above?

Binomial distributions examples

Solving using Binomial distributions: Over Booking example, Solution

- $X \sim B(32, 0.8)$ which means $n = 32$ and $P(\text{show}) = 0.8$, $p(\text{no show}) = 0.2$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

To solve this problem, what do you need to find using the above?

$$\begin{aligned} P(\text{Overbooked}) &= P(X = 32) + P(X = 31) \\ &= \binom{32}{32} 0.8^{32} (0.2)^{32-32} + \binom{32}{31} 0.8^{31} (0.2)^{32-31} \\ &= 0.8^{32} + (32 * 0.8^{31} * 0.2) \\ &= 0.00713 \end{aligned}$$

Binomial distributions examples

Key takeaways

Discrete random variable:

- mean $\mu = \sum x_i P(X = x_i)$
- variance $\sigma^2 = \sum (x_i - \mu)^2 P(X = x_i)$
- standard deviation $\sigma = \sqrt{\sigma^2}$

Binomial random variable $X \sim B(n, p)$

- Probability distribution

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- mean $\mu = np$
- variance $\sigma^2 = np(1-p)$